ALPHA SYSTEM REPLACEMENT

GDPS Hardware Acquisition Statement of Work

CDW-G Technical Validation and Budgetary Quote Draft

Project	**PR-26-GPO-2540 — Alpha System Replacement**
System	GDPS
Acquisition Type	Commercial hardware, storage, licensing, support, and implementation services
Preferred Reseller	CDW Government LLC (CDW-G)
Classification	UNCLASSIFIED
Status	Technical Validation / Budgetary Quote Draft v1.3 - Cross-Data-Center Failover

Purpose: obtain an engineered, validated, fully supportable bill of materials and budgetary quote for the eight dedicated GDPS physical file/service cluster nodes and the TEST, PROD, DR, and COOP shared-storage infrastructure.

Executive Summary

The U.S. Government Publishing Office is replacing the legacy Alpha/OpenVMS hardware platform supporting congressional publishing workflows with GDPS. The replacement preserves the existing clustered file-service, shared-storage, replication, failover, and COOP recovery capabilities while moving the platform to supported modern infrastructure.

This acquisition package requests CDW-G/HPE technical validation of the proposed physical file/service cluster nodes, shared storage, Fibre Channel, licensing, support, and implementation requirements before final procurement. TEST is intentionally designed to mirror PROD. DR replaces and improves the ALPHA6 Production Backup capability, and COOP preserves the DCIAC recovery model.

The dedicated physical-server acquisition baseline is eight (8) servers: two TEST file/service cluster nodes, two PROD file/service cluster nodes, two DR file/service cluster nodes, and two COOP file/service cluster nodes. DEV is intentionally implemented as four virtual machines on the existing GPO virtualization platform and does not require dedicated physical server acquisition. The solution also includes production-capable shared storage for TEST, PROD, DR, and COOP, with TEST/PROD/DR at a minimum of 14.4 TB usable each and COOP at a minimum of 21.4 TB usable.

GDPS Hardware Acquisition Statement of Work - Physical Lift-and-Shift Baseline

GDPS is a lift-and-shift modernization of the existing Alpha estate. The confirmed two-node physical cluster pattern is preserved for TEST, PROD, DR, and COOP. DEV is the explicit efficiency exception and is hosted as virtual machines on existing GPO virtualization infrastructure. This SOW does not authorize silent substitution of the eight physical cluster nodes with VMs.

VALIDATION INTENT: This draft is not a final manufacturer-part-number order. CDW-G/HPE should convert the requirements below into a complete engineered BOM with exact manufacturer SKUs, CDW-G SKUs, quantities, support, replication licensing, optics/cabling, implementation services, and contract pricing.

Acquisition Element	Baseline Quantity/Capacity	Purpose / Validation
DEV hosting	4 VMs - no dedicated physical purchase	Existing GPO virtualization platform; external HQ NAS for DEV file data
TEST physical file/service nodes	2	Dedicated two-node WSFC; production mirror
PROD physical file/service nodes	2	Dedicated two-node WSFC; authoritative production file service
DR physical file/service nodes	2	Dedicated two-node WSFC; replaces/improves ALPHA6 Production Backup / DR
COOP physical file/service nodes	2	Dedicated two-node WSFC in DCIAC/CRATE recovery environment
TOTAL DEDICATED PHYSICAL SERVERS	8	Physical acquisition baseline
TEST shared storage	14.4 TB usable minimum	Same architecture/class and behavior as PROD
PROD shared storage	14.4 TB usable minimum	Authoritative production storage; replication source
DR shared storage	14.4 TB usable minimum	Production-recovery target
COOP shared storage	21.4 TB usable minimum	DCIAC recovery storage; receives PROD replication
Fibre Channel fabrics	Proposed 2 per TEST/PROD/DR/COOP if dedicated fabric required	CDW-G to validate reuse vs new acquisition
Server FC HBAs	Redundant for all 8 physical nodes	CDW-G to validate current-generation supported adapter design
Support	5-year preferred	HPE Tech Care / storage support / DMR where applicable
Implementation services	Required	Installation, configuration, integration, replication, cluster support, documentation

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Section B - Supplies or Services and Prices/Costs

Final CLIN structure and funded amounts will be established by GPO Acquisition Services. The line items below are the recommended technical grouping for CDW-G budgetary validation and quote development.

Item	Supplies / Services	Qty	Unit	Price	Description
0001	TEST physical cluster + storage infrastructure	1	Lot	TBD by CDW-G	Two dedicated physical WSFC file/service nodes, >=14.4 TB usable shared storage, FC connectivity, support, optics/cables and implementation. TEST mirrors PROD.
0002	PROD physical cluster + storage infrastructure	1	Lot	TBD by CDW-G	Two dedicated physical WSFC file/service nodes, >=14.4 TB usable shared storage, redundant connectivity, support and replication source capability.
0003	DR physical cluster + storage infrastructure	1	Lot	TBD by CDW-G	Two dedicated physical WSFC file/service nodes, >=14.4 TB usable recovery storage, redundant connectivity, support and PROD recovery replication target capability.
0004	COOP physical cluster + storage infrastructure	1	Lot	TBD by CDW-G	Two dedicated physical WSFC file/service nodes, >=21.4 TB usable DCIAC/CRATE storage, redundant connectivity, support and replication target capability.
0005	Software/licensing and support not bundled in hardware CLINs	1	Lot	TBD by CDW-G	Windows Server licensing for eight physical nodes if not covered by GPO enterprise licensing; HPE replication software; management licensing; five-year support.

0006	Professional services	1	Lot	TBD by CDW-G	Engineering validation, rack/stack, firmware baseline, SAN zoning support, cluster/storage configuration, replication configuration, knowledge transfer and as-built documentation.

Section C - Description / Specifications / Statement of Work

1. Background

GPO operates a legacy Alpha/OpenVMS environment that supports congressional publishing and related file-processing workflows. The existing environment uses clustered compute, shared HPE 3PAR/Primera storage, Fibre Channel connectivity, data replication, production failover, and a self-contained COOP Data Center in a Crate. The Alpha platform is being replaced by GDPS.

The replacement must preserve the existing operational availability and file-service behavior required by current applications while moving to supported modern technology. TEST is being added as a new pre-production environment so the production cluster/storage/replication topology can be fully exercised before deployment.

2. Scope

PHYSICAL LIFT-AND-SHIFT BASELINE: This SOW intentionally prices dedicated physical servers because the existing Alpha environment being replaced is a physical server estate. A VM-based implementation is not an assumed optimization. If GPO Infrastructure chooses virtualization, GPO Infrastructure is responsible for that decision and for demonstrating equivalent or better performance, availability, isolation, failover, storage connectivity, security, and supportability.

GPO requires CDW-G to provide technical validation and a complete bill of materials for the eight dedicated physical file/service cluster servers, the TEST/PROD/DR/COOP shared-storage infrastructure, connectivity, licensing, support, and implementation services required by the GDPS lift-and-shift design.

Physical server naming is architecture-controlled. The eight servers shall be referenced in engineering, quote, implementation, support, and as-built documentation by these exact names: HQVMGDPSFST1, IMVMGDPSFST2, HQVMGDPSFSP1, IMVMGDPSFSP2, IMVMGDPSFSD1, IMVMGDPSFSD2, IMVMGDPSFSC1, and IMVMGDPSFSC2. CDW-G shall not substitute alternate hostnames in its design package.

Cross-data-center resiliency is a mandatory acquisition requirement. HQ and IM are separate data centers. TEST shall use HQVMGDPSFST1 at HQ and IMVMGDPSFST2 at IM; PROD shall use HQVMGDPSFSP1 at HQ and IMVMGDPSFSP2 at IM. CDW-G/HPE shall validate that the proposed server, Fibre Channel, storage, replication, and support configuration can sustain service failover to the surviving data center when either HQ or IM is unavailable.

- Eight (8) dedicated physical file/service cluster servers total: two TEST, two PROD, two DR, and two COOP. DEV uses four VMs on existing GPO virtualization infrastructure.
- Four (4) enterprise shared-storage systems/allocations: TEST, PROD, DR, and COOP.

- TEST and PROD must be architectural mirrors at the file-cluster and storage layer. HQ and IM are separate GPO data centers; each TEST and PROD cluster shall place one physical node at HQ and its peer node at IM.
- PROD storage must support recovery replication to DR and COOP storage.
- TEST must support the same replication/failover mechanics used by PROD so the production design can be rehearsed.
- DR and COOP must remain independently recoverable from a primary production outage; COOP must remain a self-contained DCIAC recovery platform.
- All required Fibre Channel adapters, switches or fabric ports, optics, cables, rails, power supplies, support, and licensing must be included in the validated solution.

BRAND / COMPATIBILITY: The current Alpha production and COOP storage environment is HPE. GPO's proposed replacement baseline is HPE server/storage infrastructure. CDW-G may recommend updated HPE models/SKUs where necessary for availability, TAA compliance, supportability, or lifecycle reasons, but substitutions must be clearly identified and technically justified.

3. Applicable Documents

Document	Purpose
GDPS Hardware Infrastructure Master Design	Defines DEV/TEST/PROD/COOP server, storage, cluster, and replication baseline
GDPS Windows File Cluster and Legacy SMB Namespace Standard	Defines WSFC, File Share Witness, legacy SMB namespace preservation, and cutover constraints
GDPS Production Cutover Operations Runbook	Defines operational switchover, validation, stop conditions, and rollback
Legacy Alpha storage diagrams	Defines current clustered 3PAR/Primera and replication baseline
CDW-G technical validation workbook	Initial acquisition categories and costing structure

4. Specific Tasks

1. Task 1 - Review the GDPS hardware, cluster, storage, and replication requirements and conduct an engineering validation.
2. Task 2 - Provide a complete HPE bill of materials with exact manufacturer part numbers and CDW-G part numbers.
3. Task 3 - Validate CPU, memory, local boot storage, NIC, redundant power, rail, firmware, operating-system compatibility, support, and redundant FC HBA configuration for all eight physical GDPS file/service cluster nodes.
4. Task 4 - Engineer TEST, PROD, and DR storage configurations that provide at least 14.4 TB usable capacity each. TEST must use the same storage architecture/class and behavior as PROD; DR must be production-recovery capable.
5. Task 5 - Engineer COOP storage providing at least 21.4 TB usable capacity in the recovery crate.
6. Task 6 - Validate required Fibre Channel fabric design, including whether existing GPO fabrics can be reused or whether new dual-fabric switches are required for each environment.
7. Task 7 - Include all required optics, transceivers, cables, licenses, rails, power components, rack accessories, and management components.
8. Task 8 - Validate HPE replication licensing and configuration required for PROD-to-DR and PROD-to-COOP recovery replication, with TEST able to rehearse the same production mechanisms.
9. Task 9 - Validate Windows Server / WSFC licensing requirements and identify licenses covered by or additional to GPO enterprise agreements.
10. Task 10 - Confirm that DEV physical server procurement is not required because DEV will run as four VMs on existing GPO virtualization infrastructure. Non-file-service application/database hosting outside DEV is not part of the eight-server physical cluster-node count in this SOW and shall not be silently added to the physical BOM without GPO direction.
11. Task 11 - Provide five-year support options for all quoted HPE hardware and software, including defective-media retention where available/appropriate.
12. Task 12 - Quote implementation/professional services and provide an as-built package after installation.

Task 13 - Validate the cross-data-center TEST and PROD design for complete loss of HQ and complete loss of IM, including server reachability, storage/data availability, quorum/witness behavior, Fibre Channel/network dependencies, and recovery/failover prerequisites.

5. Detailed Technical Requirements

5.1 Physical File-Cluster Nodes

Environment	Qty	Function	Special Requirement
TEST	2	Windows Server Failover Cluster file/service nodes	Physical pair; TEST mirrors PROD Physical names: HQVMGDPSFST1 / IMVMGDPSFST2.
PROD	2	Windows Server Failover Cluster file/service nodes	Physical production pair Physical names: HQVMGDPSFSP1 / IMVMGDPSFSP2.
DR	2	Windows Server Failover Cluster file/service nodes	Physical recovery pair replacing ALPHA6 capability Physical names: IMVMGDPSFSD1 / IMVMGDPSFSD2.
COOP	2	Windows Server Failover Cluster file/service nodes	Physical recovery pair in DCIAC/CRATE Physical names: IMVMGDPSFSC1 / IMVMGDPSFSC2.

- Rack-mount HPE enterprise servers supported for Windows Server Failover Clustering.
- Two processors or an engineered CPU configuration sized for file-service workload and future growth.
- Minimum 32 GB RAM per file-cluster node; CDW-G/HPE should recommend the supported production configuration.
- Mirrored/local boot storage sized for Windows Server, cluster tools, logs, and management software; business data must reside on shared enterprise storage.
- Redundant hot-plug power supplies.
- Redundant production Ethernet connectivity.
- Redundant Fibre Channel connectivity to independent fabric/path infrastructure.
- All required rails, cable-management hardware, power cords, transceivers, and accessories.
- TPM, Secure Boot, current firmware, and vendor-supported Windows Server configuration.
- TAA-compliant configuration where required by GPO acquisition policy.

5.2 Shared Enterprise Storage

Environment	Minimum Usable Capacity	Availability	Replication
TEST	14.4 TB usable minimum	Same platform/class and configuration approach as PROD	Production-mirror testing
PROD	14.4 TB usable minimum	Enterprise HA shared storage	Authoritative source; replicate to DR and COOP
DR	14.4 TB usable minimum	Enterprise HA recovery storage	Receives PROD recovery replication
COOP	21.4 TB usable minimum	Enterprise HA storage integrated with recovery crate	Receives PROD replication; independently operable

- Usable capacity means capacity available to GDPS after RAID/data-protection and vendor-required overhead, not raw installed capacity.
- Shared storage must support simultaneous connectivity from both nodes of the environment's file cluster.

- Redundant storage controllers and redundant host paths are required.
- Fibre Channel is the preferred baseline because it most closely preserves the current Alpha production storage pattern; CDW-G may recommend the current HPE supported design.
- The storage design must support Windows MPIO or the vendor-supported equivalent multipath configuration.
- The storage solution must support snapshots, monitoring, firmware management, and replication required by the GDPS recovery design.

5.3 Fibre Channel Connectivity

- Two independent storage paths/fabrics are required for TEST, PROD, DR, and COOP. For TEST and PROD, the storage design must also support the cross-data-center node placement and continued/recoverable data access after loss of either HQ or IM.
- Each physical file-cluster node must have redundant FC connectivity with no single HBA/path dependency.
- CDW-G shall validate whether existing GPO Brocade/HPE SAN fabric capacity can be reused.
- If new switches are required, quote a redundant pair per environment and include the correct licenses, SFPs/transceivers, optics, cables, support, and management requirements.
- Provide a port-count and connectivity diagram as part of the validation package.

5.4 Replication

- PROD shared storage must replicate the required GDPS dataset to DR and COOP recovery storage.
- TEST must use the same replication technology/mechanism as PROD so failover and recovery can be rehearsed.
- Quote all replication software licenses, array licenses, management components, and support required for the complete design.
- Identify any bandwidth, latency, or network dependencies needed to meet the replication design.

5.5 File Share Witness

TEST, PROD, DR, and COOP use a File Share Witness for WSFC quorum. The witness does not require dedicated array capacity and must not reside on clustered business-data storage. For TEST and PROD, the witness must not be dependent exclusively on HQ or IM and must remain reachable from the surviving data center during a site outage. No dedicated witness hardware is required unless GPO determines that existing independent witness hosting is unavailable.

5.6 DEV Virtual Machine Baseline and Physical Acquisition Boundary

Environment	Dedicated Physical Roles	Baseline
DEV	4 VMs: Client/UI, Middleware, File Processor, PostgreSQL	No dedicated physical server purchase
TEST	2 physical file/service cluster nodes	Dedicated physical pair; other logical roles are not counted in this physical BOM Physical names: HQVMGDPSFST1 / IMVMGDPSFST2.
PROD	2 physical file/service cluster nodes	Dedicated physical pair; other logical roles are not counted in this physical BOM Physical names: HQVMGDPSFSP1 / IMVMGDPSFSP2.
DR	2 physical file/service cluster nodes	Dedicated physical recovery pair Physical names: IMVMGDPSFSD1 / IMVMGDPSFSD2.
COOP	2 physical file/service cluster nodes	Dedicated physical DCIAC/CRATE recovery pair Physical names: IMVMGDPSFSC1 / IMVMGDPSFSC2.

DEV Client/UI, Middleware, File Processor, and PostgreSQL roles shall be provisioned as four VMs on the existing GPO virtualization platform. They are not part of the CDW-G physical-server BOM. This SOW physically acquires the eight dedicated TEST/PROD/DR/COOP file/service cluster nodes. Any additional physical application/database servers require separate GPO direction and are not to be assumed by CDW-G.

5.7 Software and Licensing

Software / License	Scope	Validation Requirement
Windows Server	Eight (8) dedicated physical file/service cluster nodes	Validate edition/core licensing and GPO enterprise agreement coverage for the eight physical nodes
Windows Failover Clustering	TEST/PROD/DR/COOP	Validate included rights/features and implementation prerequisites
HPE storage replication	PROD->DR and PROD->COOP; TEST equivalent rehearsal	Quote all required replication licenses/features
HPE management	Servers/storage	Include management licenses/subscriptions required for full support and monitoring
DEV virtualization	Four DEV VMs on existing GPO virtualization infrastructure	No dedicated physical DEV server acquisition

5.8 Support and Services

- Preferred support term: five (5) years for production hardware and storage.
- Provide HPE Tech Care or current equivalent appropriate for 24x7 production use.
- Include defective-media retention where available and appropriate for federal data handling.
- Provide firmware/driver baseline recommendations.
- Provide installation/rack-and-stack support if required.
- Provide storage configuration, SAN integration, replication configuration, and knowledge transfer.
- Provide final as-built diagrams, port maps, serial/asset inventory, firmware baseline, support contract identifiers, and configuration documentation.

6. Proposed Acquisition Baseline / BOM Categories

Category	Baseline Quantity	Notes
DEV virtual machines	4	Internal GPO virtualization provisioning; not CDW-G physical purchase
WSFC physical file/service servers	8	2 TEST + 2 PROD + 2 DR + 2 COOP
Enterprise shared-storage systems/allocations	4	TEST + PROD + DR + COOP
Dual-port FC HBAs	16 proposed minimum	2 per physical cluster server proposed; CDW-G to validate exact design
FC switches/fabrics	8 proposed if dedicated	2 per TEST/PROD/DR/COOP; validate reuse of existing SAN first
FC optics/transceivers	TBD	Size to validated HBA/switch/storage port design
FC cables	TBD	Size/length per rack/fabric design
Ethernet adapters/optics/cables	TBD	Validate redundant network design for all 8 physical servers
Windows Server licenses	Up to 8 physical servers	Quote only incremental licenses required after GPO enterprise entitlements
HPE replication licensing	1 complete solution	PROD->DR + PROD->COOP + TEST equivalent rehearsal capability
Five-year HPE server support	8 servers	Role-appropriate HPE Tech Care / equivalent
Five-year infrastructure support	Storage + FC	Arrays, switches/software as applicable
Professional services	1 lot	Engineering, installation, integration, documentation, knowledge transfer

7. CDW-G / HPE Technical Validation Requirements

Validation Item	Required?	CDW-G Response
Exact HPE server model/SKU	Required	Provide complete configured BOM for each of eight nodes
CPU/memory sizing	Required	Confirm recommended production configuration and TEST mirror
Boot/local disk layout	Required	Windows/cluster software only; no authoritative business-data storage
FC HBA model/count	Required	Redundant paths with correct supported firmware/drivers
Storage model/configuration	Required	Usable capacity, RAID/data protection, controllers, drives, cache, ports
TEST/PROD equivalence	Required	Confirm identical storage architecture/class and file-cluster-node configuration
COOP crate compatibility	Required	Confirm rack/power/thermal/space and independent operation
SAN switch/port design	Required	Confirm reuse vs new switches, port licenses, optics, zoning requirements
Replication licensing	Required	Identify all features/licenses and compatibility among PROD, DR, COOP, and TEST rehearsal
Windows licensing	Required	Identify incremental license requirement after GPO enterprise entitlements
Support	Required	5-year options and SLA details
Implementation services	Required	Scope, assumptions, exclusions, estimated effort
TAA / federal compliance	Required	Identify compliant SKUs and any exceptions
Lifecycle	Required	No end-of-sale/end-of-support components in proposed design
Physical-server quantity/topology	Required	Confirm eight dedicated physical servers: 2 TEST, 2 PROD, 2 DR, 2 COOP; DEV is virtualized
HQ/IM site-loss resilience	Required	Validate node placement and all dependencies for TEST/PROD failover when either HQ or IM data center is unavailable.

NO SILENT SUBSTITUTIONS: If CDW-G/HPE proposes a different server, storage, switch, HBA, or software product than the baseline assumption, the quote must identify the substitution and explain the technical reason, compatibility impact, support lifecycle, and whether TEST/PROD parity is preserved.

8. Deliverables and Delivery Schedule

#	Deliverable	Due	Minimum Content
1	Engineering validation report	Prior to final quote	Validated architecture, exceptions, assumptions, dependencies
2	Complete manufacturer BOM	With final quote	HPE part number, CDW-G part number, description, quantity, environment
3	Budgetary/formal quote	With BOM	Unit price, extended price, contract discount, support, services
4	Storage usable-capacity worksheet	With final quote	Raw vs usable capacity and protection overhead
5	Connectivity/port map	With final quote	FC/Ethernet ports, HBAs, switches, optics, cables
6	Licensing matrix	With final quote	Windows licensing for 8 physical servers, HPE

			replication, management, and confirmation that DEV uses existing GPO virtualization
7	Support matrix	With final quote	Coverage period, service level, DMR, support identifiers
8	Implementation plan	Before installation	Rack/stack, storage, SAN, replication, validation activities
9	As-built documentation	At completion	Final diagrams, serials/assets, firmware, configuration, port map

Requested delivery date: to be established by GPO Acquisition Services after CDW-G validates lead times. CDW-G shall provide current estimated ship dates and identify long-lead components in the quote.

9. Government-Furnished Equipment and Information

- GPO will provide rack/location information and available power information.
- GPO will provide existing SAN/fabric information required to determine whether existing Brocade/HPE fabric can be reused.
- GPO Infrastructure will provide the approved physical-server naming, rack, power, network, and data-center placement standards required to finalize the eight-server physical cluster-node build. Network/IP and security-boundary details are governed separately by GPO Network and Security teams.
- GPO will provide final network addressing, DNS, AD, witness location, and security configuration through the appropriate infrastructure teams.
- GPO will provide the GDPS hardware master design, SMB namespace standard, and cutover runbook for integration context.

10. Place of Performance

Engineering and quote development may be performed at the vendor's facility. Hardware will be delivered to GPO-designated locations. COOP crate components must be compatible with the designated recovery-crate location and constraints.

11. Security

The acquisition and hardware are UNCLASSIFIED. Vendor access to GPO systems, facilities, or configuration data is subject to GPO security and access requirements.

Section D - Packaging and Marking

- All equipment shall be new, manufacturer-supported, and shipped in original manufacturer packaging.
- Quote and packing documentation shall identify GPO project/PO information as provided by Acquisition Services.
- Hardware shall be clearly identifiable by manufacturer part number and serial number.
- COOP components shall be identified as COOP/CRATE where practical to prevent misallocation.

Section E - Inspection and Acceptance

Acceptance Area	Acceptance Requirement
Hardware count	All quoted items received in correct quantities
Model/SKU validation	Received manufacturer part numbers match the accepted BOM or formally approved substitutions
Physical condition	No shipping damage; all components new and serviceable

Support entitlement	All servers/storage/switches/software registered with correct support term
Firmware/supportability	Configuration supported by HPE and compatible with required Windows/WSFC/storage stack
Storage capacity	Usable capacity meets or exceeds 14.4 TB TEST, 14.4 TB PROD, 14.4 TB DR, 21.4 TB COOP
Connectivity	Required redundant FC/network components delivered
Documentation	As-built and support information delivered

Section F - Delivery or Performance

CDW-G shall provide the estimated lead time for each major hardware component and identify any item that could delay the overall solution. Partial delivery may be accepted only when coordinated with GPO and when it does not create configuration-control or asset-tracking issues.

Environment	Delivery Location	Delivery Requirement
TEST	GPO-designated TEST infrastructure location	Complete production-mirror hardware set
PROD	GPO-designated production infrastructure location	Complete production clustered file/storage set
COOP	GPO-designated COOP crate location	Complete recovery clustered file/storage set
DR	GPO-designated DR / IMDC infrastructure location	Complete physical DR clustered file/storage set

Appendix A - Environment Hardware Matrix

Environment	Dedicated Physical Servers	Shared Storage	Cluster	Quorum	Special Requirement
DEV	0 physical; 4 VMs	2 TB external HQ NAS allocation	No dedicated physical file cluster	N/A	Virtualized development stack
TEST	2 physical file/service nodes	14.4 TB usable minimum	2-node physical WSFC	File Share Witness	Mirrors PROD cluster/storage behavior Physical names: HQVMGDPSFST1 / IMVMGDPSFST2.
PROD	2 physical file/service nodes	14.4 TB usable minimum	2-node physical WSFC	File Share Witness	Authoritative production; replicates to DR and COOP Physical names: HQVMGDPSFSP1 / IMVMGDPSFSP2.
DR	2 physical file/service nodes	14.4 TB usable minimum	2-node physical WSFC	File Share Witness	Production Backup / DR replacement Physical names: IMVMGDPSFSD1 / IMVMGDPSFSD2.
COOP	2 physical file/service nodes	21.4 TB usable minimum	2-node physical WSFC	File Share Witness	Self-contained DCIAC/CRATE recovery Physical names: IMVMGDPSFSC1 / IMVMGDPSFSC2.

Appendix B - CDW-G Quote Response Template

Quote Field	CDW-G Response
Manufacturer	
Manufacturer Part Number	
CDW-G Part Number	
Description	
Environment (TEST/PROD/DR/COOP/Shared)	
Quantity	
Unit Price	
Extended Price	
Contract Discount / Vehicle	
Support SKU / Term	
Lead Time	
TAA Compliant	Yes / No / N/A
Dependency / Required Accessory	
Substitution from Baseline	Yes / No
Technical Notes	